



SEEK WISDOM, ELEVATE YOUR INTELLECT AND SERVE HUMANITY !

Addis Ababa University
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PLACENTAL ABRUPTION

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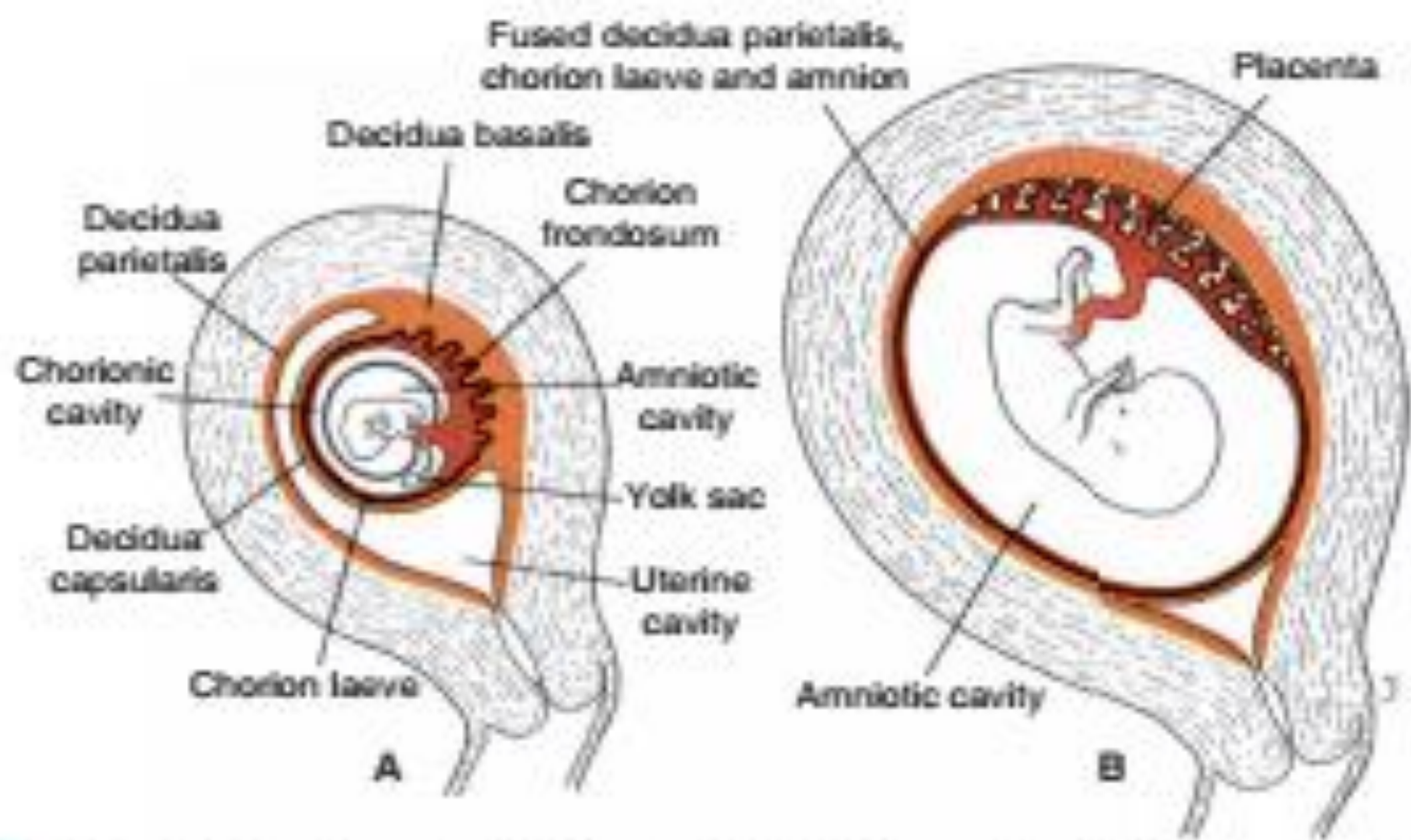


Figure 8.10 Relation of fetal membranes to wall of the uterus. A. End of the second month. Note the yolk sac in the chorionic cavity between the amnion and chorion. At the abembryonic pole, villi have disappeared (chorion laeve). B. End of the third month. The amnion and chorion have fused, and the uterine cavity is obliterated by fusion of the chorion laeve and the decidua parietalis.

OUTLINE

- Introduction
- Etiopathogenesis
- Risk factors
- Classification
- Diagnosis
- Management

INTRODUCTION

- In the 2nd half of px, **3-placental disorders** contribute substantially to maternal mortality rates.
- These include
 - Placental abruption,
 - Placenta previa, and
 - Placenta accreta spectrum

DEFINITION

- Partial/total separation of the placenta from its implantation site **before delivery**.
- Premature separation of the **normally implanted** placenta (unlike pp)

ETIOPATHOGENESIS

- The immediate cause of the premature placental separation is **rupture of maternal vessels in the decidua basalis**.
 - Rarely, the bleeding originates from the fetal-placental vessels.
- The accumulating blood splits the decidua, separating off a thin layer of decidua with its placental attachment.

RISK FACTORS

① MATERNAL FACTORS	③ PLACENTAL /AF FACTORS	④ FETAL FACTORS
• AMA • Multiparity • Cigarette smoking (2*) • Cocaine use—(16*)? • Myoma	• Prior abruption (5.8—22%) • Sub-chorionic hematoma • Hydramnios • PROM (5-17%) • IAI	• Multiple gestation • Prematurity(LBW) ? • Single umbilical artery
⑤ OTHERS • HDP –Gestational HTN –Chronic HTN—1% –Preeclampsia syndrome—4% • INFECTIONS –UTI, IAI • HYPOTHYROIDISM –Subclinical hypothyroidism /↑levels of anti-thyroid Abs is associated with a 2—3* higher risk for AP. [ACOG 2020] • No convincing evidence supports a link b/n thrombophilias & placental abruption.		

Risk Factor	Relative Risk
Prior abruption	10–188
Increased age and parity	1.3–2.3
Preeclampsia	2.1–4.0
Chronic hypertension	1.8–3.0
Chorioamnionitis	3.0
Preterm ruptured membranes	2.4–4.9
Multifetal gestation	2–8
Low birthweight	14.0
Hydramnios	1.9
Cigarette smoking	1.4–1.9
Single umbilical artery	3.4
Cocaine use	—
Uterine leiomyoma	2.6
Subchorionic hematoma	5.7

Risk of recurrence

- ❖ In a population based study of 767,000 pregnancies:
 - **7*** higher risk for recurrence of a “mild” abruption and
 - **12*** risk for a “severe” abruption (Rasmussen, 2009).
 - Abruptio & IUFD have a recurrence rate of **12%**,
 - Half of these AP caused another fetal death
 - **50*** risk for women who had 2-severe abruptions.

Cont'd...

- Management of a pregnancy subsequent to an abruption is difficult
 - because **another separation may suddenly occur**, even remote from term.
- In many of these recurrences, evaluation of fetal well-being is **almost always reassuring** beforehand. Thus, **antepartum fetal testing** is usually **not predictive**.
- Because **term abruptions tend to be recurrent**, labor induction at **37 weeks** is recommended.

The frequency of placental abruption rises with advanced maternal age

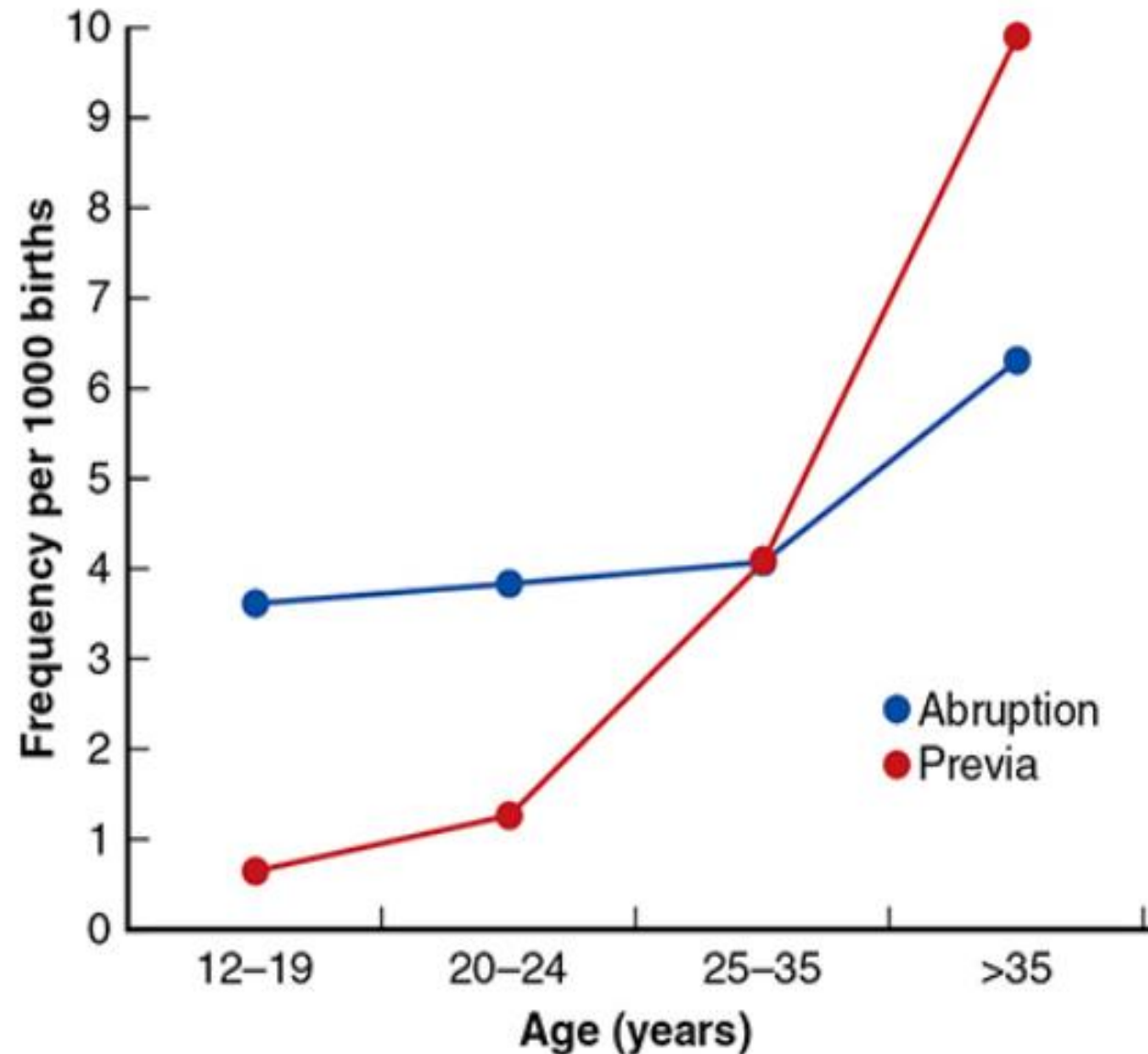


FIGURE 43-3 Frequency of placental abruption and placenta previa by maternal age at Parkland Hospital from 2000 through 2015.

The frequency of placental abruption **ris**es across the 3rd TM

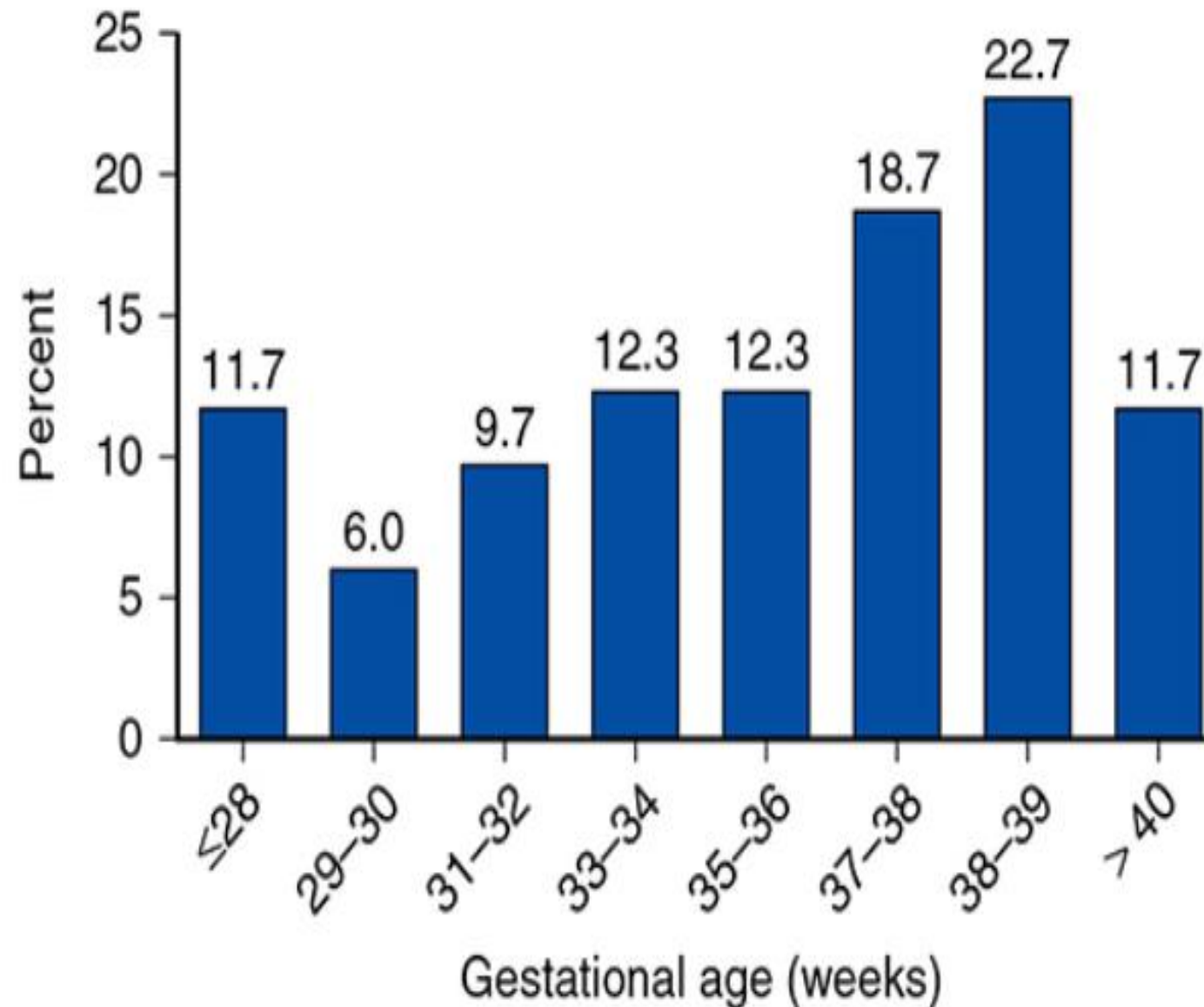


FIGURE 43-4 Frequency of placental abruption by gestational age at Parkland Hospital.

CLASSIFICATION

- Classification of placental abruption is based on:
 - Temporal classification
 - Extent of separation
 - Location of separation
 - Severity of sx

Cont'd

Acute abruption

- High-pressure **arterial hemorrhage** in the central area of the placenta extensively dissects through the placental-decidual interface → Placental separation.

Chronic abruption

- Early abnormal spiral artery development →decidual necrosis, placental inflammation (and possibly infarction), and ultimately low-pressure **venous hemorrhage**, typically at the periphery of the placenta (ie, marginal abruption).
- Light intermittent bleeding, oligohydramnios, and FGR

CONT'D

Extent of separation

- Partial vs complete

Location of separation

- Marginal vs central

CONT'D

- Class 0 – Asymptomatic, a dx of class 0 is made retrospectively
 - By finding of an **organized clot/ depressed area** on a delivered placenta.
- Class 1 - Mild (represents ~**48%** of all cases)
- Class 2 - Moderate (represents ~**27%** of all cases)
- Class 3 - Severe (represents ~**24%** of all cases)

CLASSIFICATION OF ABRUPTIO PLACENTAE

			GRADE-1(MILD)	GRADE-2(MODERATE)	GRADE-3(SEVERE)
CLINICAL FINDINGS	MATERNAL CONDITION	1. BLOOD LOSS(In ml)	<400ml (No—mild v-bleeding)	400-1000ml (No—moderate v-bleeding)	>1000ml (No—severe v-bleeding)
		2. BP(SHOCK)	Normal BP	Orthostatic changes in BP	Maternal shock
		3. PR	Normal HR	↑HR	↑HR
		4. UTERINE IRRITABILITY (PAIN/TENDERNESS)	Normal or Slightly tender uterus	Moderate to severe uterine tenderness	Very painful tetanic uterus
	FETAL CONDITION	FHB	Normal	Abnormal	Fetal distress/death
LAB FINDINGS	FIBRINOGEN LEVEL(mg/dl)		Normal	150-250	<150 mg

DIAGNOSIS

- **Hx:**
 - **Vaginal bleeding-----(78%)**
 - Abdominal pain
- **PE:**
 - abdominal tenderness---(66%)

- **Lab:**
 - CBC, BG & Rh
 - Coagulation profile
 - PT, PTT,
 - Fibrinogen or
 - bedside bleeding & clotting.

- **US:**
 - Fetal assessment: NR-BPP-----(60%)
 - Retroplacental clot and
 - For exclusion of placenta previa

MANAGEMENT

- BLS protocol

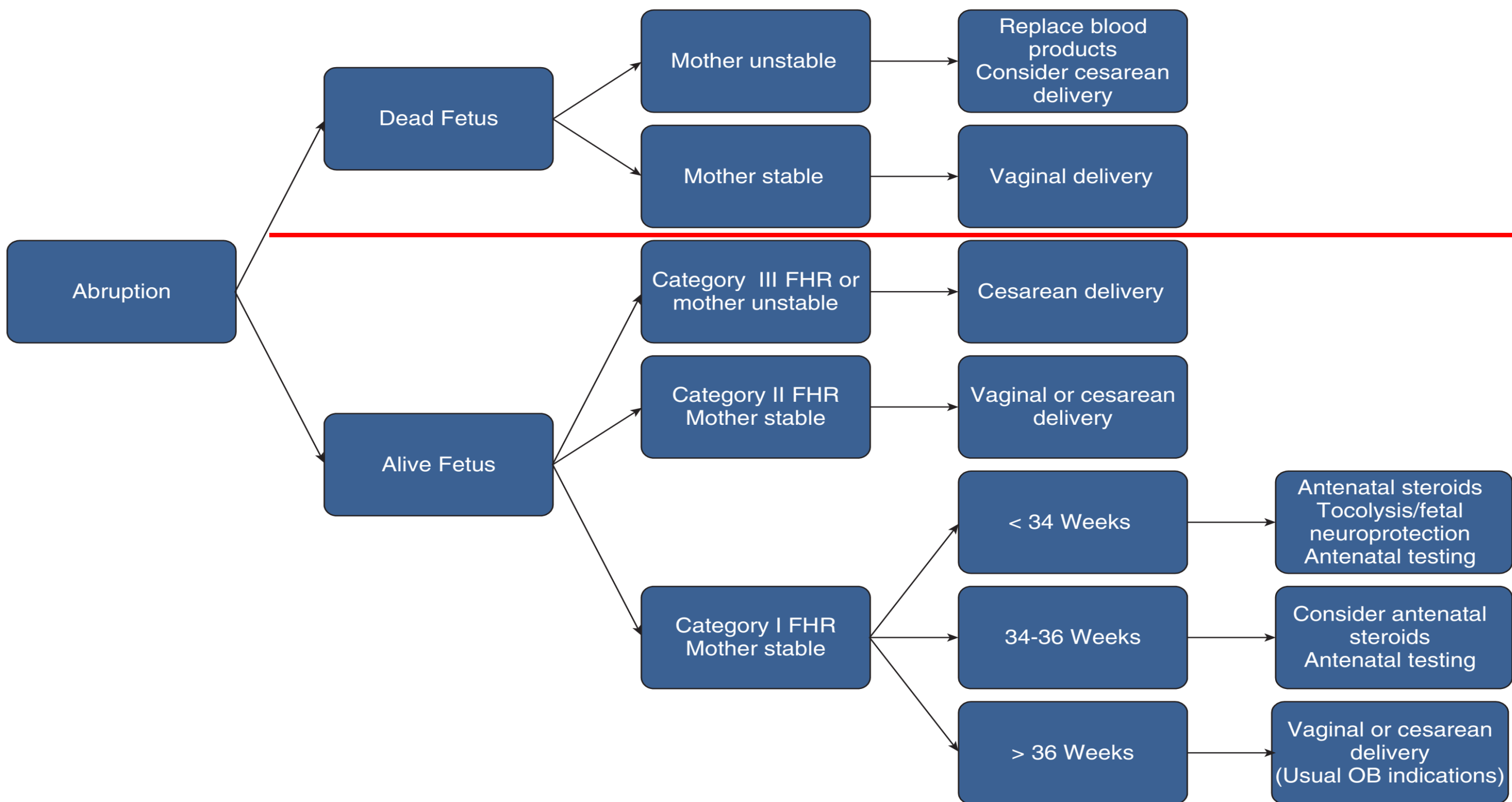


Fig. 18.4 Algorithm for Management of Abruptio. *FHR*, Fetal heart rate; *OB*, obstetric.

Abruptio placentae	SUMMARY
Definition	• Placental detachment from the uterus before fetal delivery
Risk factors	• Hypertension, preeclampsia • Abdominal trauma • Prior abruptio placentae • Cocaine & tobacco use
Clinical presentation	• Sudden-onset vaginal bleeding • Abdominal or back pain • High-frequency, low-intensity contractions • Rigid, tender uterus
Diagnosis	• Clinical • Ultrasound: ± Retroplacental hematoma
Complications	• Fetal hypoxia, preterm birth, mortality • Maternal hemorrhage, disseminated intravascular coagulation

REFERENCES

- Williams Obstetrics 26th edition
- Gabbe's Obstetrics 9th edition

THANK YOU